Question:

Let \(O=(0,0)\), \(A=\left(\tfrac{1}{2},0\right)\), and \(B=\left(0,\tfrac{\sqrt{3}}{2}\right)\) be points in the coordinate plane. Let \(\mathcal{F}\) be the family of segments \(\overline{PQ}\) of unit length lying in the first quadrant with \(P\) on the \(x\)-axis and \(Q\) on the \(y\)-axis. There is a unique point \(C\) on \(\overline{AB}\), distinct from \(A\) and \(B\), that does not belong to any segment from \(\mathcal{F}\) other than \(\overline{AB}\). Then \(OC^2=\tfrac{p}{q}\), where \(p\) and \(q\) are relatively prime positive integers. Find \(p+q\).

Reasoning + Answer:

<think> To solve this problem, we start by understanding the geometry of the problem and the conditions given. We have points \(O=(0,0)\), \(A=\left(\frac{1}{2},0\right)\), and \(B=\left(0,\frac{\sqrt{3}}{2}\right)\). The family \(\mathcal{F}\) consists of segments \(\overline{PQ}\) where \(P\) is on the x-axis and \(Q\) is on the y-axis, and the length of \(\overline{PQ}\) is 1. We need to find the point \(C\) on \(\overline{AB}\) that does not belong to any segment in \(\mathcal{F}\) except \(\overline{AB}\) itself.

First, we parameterize the line \(\overline{AB}\). The line \(\overline{AB}\) can be described by the equation:
\[y = \frac{\sqrt{3}}{2} x \]
Since \(C\) is on \(\overline{AB}\), we can write the coordinates of \(C\) as \(C = (x_0, \frac{\sqrt{3}}{2} x_0)\). The distance from \(C\) to the axes is \(x_0\) and \(\frac{\sqrt{3}}{2} x_0\). Since \(P\) and \(Q\) lie on the axes and form a segment of length 1, the coordinates of \(P\) are \((1, 0)\) and the coordinates of \(Q\) are \((0, 1)\) (since the distance from the origin to \(P\) is 1 and the distance from the origin to \(Q\) is 1, and the segments are of unit length and lie in the first quadrant).

For \(C\) to be the unique point that does not belong to any segment in \(\mathcal{F}\) except \(\overline{AB}\), \(C\) must be the point such that the distance from \(C\) to the x-axis and y-axis does not allow any other segment in \(\mathcal{F}\) to pass through it other than \(\overline{AB}\). This unique point \(C\) is the point that lies exactly in the middle of \(\overline{AB}\). The midpoint of \(\overline{AB}\) is:
\[C = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} \right) \]
However, we need to find the distance \(OC^2\). The coordinates of \(C\) are \((x_0, \frac{\sqrt{3}}{2} x_0)\). Since \(C\) is the unique point on \(\overline{AB}\) that does not belong to any segment in \(\mathcal{F}\) other than \(\overline{AB}\), the coordinates must be exactly the midpoint of \(\overline{AB}\), and the distance from the origin to \(C\) is:
\[OC = \sqrt{ \left(\frac{1}{2} \right)^2 + \left(\frac{\sqrt{3}}{4} \right)^2 } = \sqrt{ \frac{1}{4} + \frac{3}{16} } = \sqrt{ \frac{4}{16} + \frac{3}{16} } = \sqrt{ \frac{7}{16} } = \frac{\sqrt{7}}{4} \]
Therefore, \(OC^2 = \frac{7}{16}\), and \(p = 7\) and \(q = 16\), so \(p + q = 23\).
</think>
<answer>23</answer>